

Soru 1

|     |  | (i) | (R) | (D) | (E) | (L) | (E) | (N) | (M) | (E) | (K) |
|-----|--|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
|     |  | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   |
| D   |  | 0   | 0   | 0   | 1   | 1   | 1   | 1   | 1   | 1   | 1   |
| (i) |  | 0   | (1) | 1   | 1   | 1   | 1   | 1   | 1   | 1   | 1   |
| S   |  | 0   | ↑   | 1   | 1   | 1   | 1   | 1   | 1   | 1   | 1   |
| C   |  | 0   | ↑   | 1   | 1   | 1   | 1   | 1   | 1   | 1   | 1   |
| O   |  | 0   | ↑   | 1   | 1   | 1   | 1   | 1   | 1   | 1   | 1   |
| (R) |  | 0   | 1   | (2) | 2   | 2   | 2   | 2   | 2   | 2   | 2   |
| D   |  | 0   | 1   | ↑   | 2   | 3   | 3   | 3   | 3   | 3   | 3   |
| (D) |  | 0   | 1   | 2   | (3) | 3   | 3   | 3   | 3   | 3   | 3   |
| (E) |  | 0   | 1   | 2   | 3   | (4) | 4   | 4   | 4   | 4   | 4   |
| V   |  | 0   | 1   | 2   | 3   | 4   | ↑   | 4   | 4   | 4   | 4   |
| (E) |  | 0   | 1   | 2   | 3   | 4   | 4   | (5) | 5   | 5   | 5   |
| L   |  | 0   | 1   | 2   | 3   | 4   | 5   | 5   | 5   | 5   | 5   |
| O   |  | 0   | 1   | 2   | 3   | 4   | 5   | 5   | 5   | 5   | 5   |
| P   |  | 0   | 1   | 2   | 3   | 4   | 5   | 5   | 5   | 5   | 5   |
| (E) |  | 0   | 1   | 2   | 3   | 4   | 5   | 6   | 6   | (6) | 6   |
| R   |  | 0   | 1   | 2   | 3   | 4   | 5   | 6   | 6   | 6   | ↑   |

En uzun ortak katar = İRDEEEE

Çalışma Zamanı =  $O(n * m)$

Bellek Kullanımı =  $O(n)$

Soru 2

Tablo

|   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 0 | 0 | 0 | 0  | 12 | 12 | 12 | 12 | 12 | 12 | 12 | 12 | 12 | 12 | 12 | 12 | 12 | 12 | 12 |
| 0 | 0 | 0 | 0  | 12 | 12 | 12 | 12 | 12 | 12 | 22 | 22 | 22 | 22 | 22 | 22 | 22 | 22 | 22 |
| 0 | 0 | 0 | 0  | 12 | 12 | 12 | 12 | 12 | 20 | 22 | 22 | 22 | 22 | 22 | 22 | 30 | 30 | 30 |
| 0 | 0 | 0 | 0  | 12 | 12 | 12 | 12 | 12 | 20 | 22 | 23 | 23 | 23 | 23 | 23 | 30 | 31 | 33 |
| 0 | 0 | 0 | 14 | 14 | 14 | 14 | 26 | 26 | 26 | 26 | 26 | 36 | 36 | 37 | 37 | 37 | 37 | 44 |
| 0 | 7 | 7 | 14 | 21 | 21 | 21 | 26 | 33 | 33 | 33 | 33 | 34 | 41 | 43 | 44 | 44 | 46 | 66 |
| 0 | 7 | 7 | 14 | 21 | 21 | 21 | 26 | 33 | 33 | 33 | 33 | 34 | 41 | 43 | 46 | 44 | 44 | 44 |

Tasınabilecek max değer = 44 =  $\sqrt{I_{n,w}}$

Optimal sırt çantası elemanları  $\{a, b, c, e\}$

Dinamik programlama yaklaşımı :

1)  $S_{Max} V(0) = 0$

2)  $Max V(0) = 0$

3) for  $i = 1$  to  $n$

4)  $S_{Max} V[i] = \max(S_{Max} V[i-1] + x_i, 0)$

5)  $Max V[i] = \max(Max(i-1), S_{Max} V(i))$

6) return  $Max V(n)$